

Assignment
(on FreeSQL and OneCompiler online platforms)

COURSE CODE: 05MC0108

COURSE NAME: Database Management Systems (DBMS)

Sr. No.	Questions	CO	Bloom's Level
1	Write a PL/SQL block to accept an employee's Basic Salary and calculate: - HRA = 20% - DA = 45% - TA = 8% - PF = 12% - Net Salary Display all salary components along with Net Salary.	CO1	L3 – Apply
2	Create a STUDENT table and write a PL/SQL block that accepts Roll Number and: - Calculates Total - Percentage - Assigns Grade - Displays Pass/Fail status (Pass if all subjects ≥ 35).	CO1	L3 – Apply
3	Write a PL/SQL program to generate the multiplication table of a user-entered number using: - FOR LOOP - WHILE LOOP	CO1	L3 – Apply
4	Write a PL/SQL block to accept an employee ID and: - Retrieve employee details - Calculate yearly salary - Display whether employee belongs to High Income ($> ₹600000/\text{year}$) or Normal Income.	CO1	L3 – Apply
5	Write a PL/SQL block that accepts three numbers and displays: - Largest number - Smallest number - Average - Also, check whether all numbers are equal.	CO1	L3 – Apply
6	Create a procedure that accepts Department Number and displays: - Total Employees - Average Salary - Maximum Salary Use OUT parameters.	CO2	L3 – Apply
7	Create a function that returns the annual salary of an employee ($\text{salary} \times 12$). Call the function inside a SELECT statement.	CO2	L3 – Apply
8	Create a procedure to update the salary of an employee using the employee ID. Handle the exception if the employee ID does not exist.	CO2	L4 – Analyze
9	Create a package named PKG_EMPLOYEE containing: - Function to calculate the annual salary of an employee. - Procedure to show employee information using the employee ID. Call each package object separately.	CO2	L4 – Analyze

10	Create a function that returns employee experience category based on joining date. Categories: - Less than 2 Years → Fresher 2–5 Years → Experienced - More than 5 Years → Senior Employee	C02	L3 – Apply																																								
11	Create a trigger that stores deleted employee records in EMP_HISTORY along with: - Deleted By - Deleted Date - Deleted Time	C03	L3 – Apply																																								
12	Create a BEFORE INSERT trigger that prevents insertion of students having age less than 18.	C03	L3 – Apply																																								
13	Create an AFTER UPDATE trigger that records old salary and new salary into SALARY_LOG table whenever salary is modified.	C03	L3 – Apply																																								
14	Create a trigger that automatically sets joining date to SYSDATE if no joining date is provided during insertion.	C03	L3 – Apply																																								
15	Create a trigger that prevents deleting employees belonging to HR department.	C03	L3 – Apply																																								
16	Use following “emp” MongoDB collection to perform Q.16 to Q.20. <table><tr><th>Empno</th><th>Ename</th><th>Sal</th><th>Desig</th><th>Dept</th></tr><tr><td>1</td><td>Sachin</td><td>60000</td><td>Manager</td><td>Purchase</td></tr><tr><td>2</td><td>Kohli</td><td>50000</td><td>Manager</td><td>Sales</td></tr><tr><td>3</td><td>Dhoni</td><td>30000</td><td>Clerk</td><td>Admin</td></tr><tr><td>4</td><td>Dravid</td><td>40000</td><td>Manager</td><td>Purchase</td></tr><tr><td>5</td><td>Bumrah</td><td>50000</td><td>Manager</td><td>Sales</td></tr><tr><td>6</td><td>Jadeja</td><td>20000</td><td>Clerk</td><td>Admin</td></tr><tr><td>7</td><td>Ashwin</td><td>45000</td><td>Manager</td><td>Sales</td></tr></table> Write a MongoDB query to display all employees working in the SALES department.	Empno	Ename	Sal	Desig	Dept	1	Sachin	60000	Manager	Purchase	2	Kohli	50000	Manager	Sales	3	Dhoni	30000	Clerk	Admin	4	Dravid	40000	Manager	Purchase	5	Bumrah	50000	Manager	Sales	6	Jadeja	20000	Clerk	Admin	7	Ashwin	45000	Manager	Sales	C04	L3 – Apply
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17	Write a MongoDB query to display employees whose salary is greater than 25000 and sort the result in descending order of salary.	C04	L3 – Apply																																								
18	Write a MongoDB query to display only the Employee Name, Designation, and Salary of employees working as MANAGER.	C05	L3 – Apply																																								
19	Write a MongoDB query to update the salary of employee Sachin by increasing it by 2000 using the \$inc operator.	C05	L3 – Apply																																								
20	Write a MongoDB query to count the total number of employees in each department using the Aggregation Pipeline.	C05	L4 – Analyze																																								